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Review

Natural and Semisynthetic Hydrocolloids as Functional Polymeric Materials: Structure–Function Relationships and Emerging Applications

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Abstract

Hydrocolloids comprise a diverse class of high-molecular-weight polymeric carbohydrates associated with a wide range of physicochemical and functional properties. This review provides an integrated analysis of natural hydrocolloids derived from algal (agar, alginate, carrageenan, fucoidan, laminarin, and ulvan), animal (chitin, chitosan, chondroitin sulfate, dermatan sulfate, keratan sulfate, heparin, heparan sulfate, glycogen, and hyaluronan), and plant (pectin, starch, and locust bean gum) sources, together with semisynthetic cellulose-based derivatives. Emphasis is placed on the relationship between molecular structure, charge density, sulfation patter, and branching degree, and how these parameters modulate hydration, gelation, and rheological behavior. Comparative analyses are presented, establishing structure–function interactions that link molecular characteristics to functional properties, including thickening, gelling, emulsifying, stabilizing, film-forming, and controlled-release capacities. The review also discusses the biological activities and application potential of these hydrocolloids in pharmaceutical, biomedical, and advanced material systems. In addition, emerging modification strategies, including chemical functionalization, crosslinking, and nanostructuring are discussed as tools to adjust their action and diversify their application range. Special attention is given to structure–rheology–gelation relationships and to the influence of molecular organization on mechanical strength, stability, and delivery performance. Current challenges associated with scalability, processability, reproducibility, and long-term functional stability are also critically discussed. Overall, this review provides a comprehensive structure–function perspective on hydrocolloids as sustainable and multifunctional polymeric materials, supporting their rational design and continued development in pharmaceutical sciences, biomedical engineering, and advanced material applications.

Keywords: hydrocolloids; structure–function relationship; rheology; biopolymer engineering; functional polysaccharides; biomaterials; pharmaceutical applications

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1. Introduction

Hydrocolloids constitute a broad class of high-molecular-weight hydrophilic polymers, primarily composed of polysaccharides (and in some cases proteins) that play a central role in the design of complex aqueous systems. Their functional performance arises from their ability to modulate viscosity, gelation, and interfacial behavior, enabling precise control over texture, stability, and microstructure in a wide range of applications [1]. While they are extensively used in food systems as thickening, gelling, emulsifying, and stabilizing agents, their relevance extends to pharmaceutical, cosmetic, and emerging material applications [2].

The functional performance of hydrocolloids is intrinsically linked to their chemical structure and macromolecular organization. Parameters such as monosaccharide composition, glycosidic linkages, branching degree, and charge density define their hydration behavior, intermolecular interactions, and gelation mechanisms, determining also their ability to form networks, stabilize dispersed systems, and interact with other components in complex formulations [3]. Accordingly, hydrocolloids may be classified not only according to their origin (e.g., plant, animal, algal, microbial, or semisynthetic), but also based on structural and functional attributes, including molecular architecture, charge characteristics, and mode of action in solution [4].

In applied systems, they can significantly alter the physical properties of aqueous media, forming gels, viscous solutions, or films, depending on concentration, operational conditions, and molecular structure [5]. Their capacity to bind water, control crystallization processes, prevent dehydration, and enhance physical stability underlies their widespread use in both conventional and advanced formulations. In addition, their performance at relatively low concentrations, often below 1%, makes them highly efficient functional ingredients from both technological and economic perspectives [6,7].

Beyond their traditional roles, hydrocolloids are increasingly being explored as adaptable polymeric platforms for advanced applications, including controlled delivery systems, functional coatings, and engineered materials. Their inherent biocompatibility, biodegradability, and sensitivity to chemical modification, crosslinking, and nanostructuring enable the design of tailored systems with enhanced performance. However, despite extensive research, a comprehensive understanding of how structural features translate into functional behavior across different application fields remains partially unexplained [7–9].

In this context, the present review aims to provide an integrated perspective on natural and semisynthetic hydrocolloids by establishing a structure–interaction–function rationale. By comparatively analyzing representative hydrocolloids from different sources, this work highlights the factors that define their physicochemical properties and functional behavior, as well as emerging strategies for their modification and application. Particular attention is given to challenges associated with processability, scalability, and performance optimization, with the goal of supporting the rational design and application of hydrocolloid-based systems in food, pharmaceutical, and material science contexts.

2. Materials and Methods

The review uses two major scientific databases, Scopus and PubMed. Search strings were constructed iteratively, combining terms for hydrocolloids, chemical structure, industrial applications, and bioactivity. The Scopus search used the pattern: TITLE-ABS-KEY (hydrocolloids AND (chemical structure OR detection OR prediction OR classification) AND (“industrial applications” OR “bioactivity”)).

The PubMed search mirrors Scopus TITLE-ABS-KEY syntax using equivalent field tags (“Document Title”, “Abstract”, “Author Keywords”) and Boolean operators.

Searches were limited to peer-reviewed journal articles and conference papers published in English. Pre-prints, theses and non-peer-reviewed sources were excluded to ensure methodological rigor and validated ML/DL applications.

To contextualize the current scientific relevance and technological interest associated with the hydrocolloids discussed in this review, Table 1 summarizes the publication activity and predominant application fields reported between 2020 and 2026. Overall, chitosan, starch, alginate, hyaluronan, and cellulose derivatives remain among the most extensively investigated hydrocolloids, largely due to their versatility, processability, and applicability in food, pharmaceutical, and biomedical systems. Conversely, ulvan, dermatan sulfate, and keratan sulfate remain comparatively underexplored despite increasing interest in their bioactivity and functional potential.

Table 1. Approximate publication activity and major application areas of the hydrocolloids discussed in this review *.

Hydrocolloid	Approximate Number of Publications (2020–2026)	Predominant Application Areas	Emerging Research Trends
Agar/agarose	≈8000	Electrophoresis, food gels, scaffolds	Smart hydrogels and bioprinting
Alginate	≈19,500	Encapsulation, drug delivery, tissue engineering	Injectable hydrogels and 3D bioprinting
κ-Carrageenan	≈8000	Food structuring, edible films, pharmaceutical gels	Active packaging and biomedical films
Fucoidan	≈6500	Anticancer systems, immunomodulation	Targeted therapeutics and nanocarriers
Laminarin	≈2700	Immunomodulation, wound healing	Functional bioactive coatings
Ulvan	≈1500	Tissue engineering, food packaging	Marine bioactive materials
Chitin	≈15,500	Packaging, wastewater treatment, biomedical materials	Sustainable composites and nanofibers
Chitosan	≈38,000	Drug delivery, antimicrobial films, tissue engineering	Stimuli-responsive biomaterials
Chondroitin sulfate	≈8800	Osteoarthritis therapy, regenerative medicine	Cartilage engineering
Dermatan sulfate	≈2200	Anticoagulant and regenerative systems	Bioactive extracellular matrices
Keratan sulfate	≈1500	Corneal and neural regeneration	Neuroregenerative biomaterials
Heparin	≈21,500	Anticoagulant therapy, drug delivery	Surface-functionalized biomaterials
Heparan sulfate	≈4900	Tissue engineering, signaling modulation	Glycoengineered matrices
Glycogen	≈2200	Nanocarriers, bioactive particles	Glycogen nanomedicine
Hyaluronan	≈24,000	Cosmetics, wound healing, regenerative medicine	Injectable and targeted biomaterials
Pectin	≈15,500	Food gels, nutraceuticals, delivery systems	Functional packaging and hydrogels
Starch	≈27,000	Food systems, bioplastics, encapsulation	Sustainable materials and smart films
Locust bean gum	≈3200	Thickening systems, edible films	Synergistic hydrogel systems

Table 1. *Cont.*

Hydrocolloid	Approximate Number of Publications (2020–2026)	Predominant Application Areas	Emerging Research Trends
Carboxymethylcellulose	≈12,400	Coatings, drug delivery, emulsification	Responsive hydrogels
Microcrystalline cellulose	≈15,000	Pharmaceutical excipients, composites	Nanocellulose reinforcement
Hydroxypropylcellulose	≈4300	Ophthalmic systems, tissue engineering	Flexible functional materials
Hydroxypropylmethylcellulose	≈8000	Controlled-release systems, food stabilization	Mucoadhesive biomaterials
Ethylcellulose	≈6500	Controlled drug release, packaging	Barrier films and nanocarriers

* Estimated from Scopus TITLE-ABS-KEY searches (2020–2026) using harmonized hydrocolloid/application-related keywords to focus on application-oriented hydrocolloid research.

3. Natural Hydrocolloid Sources

3.1. Hydrocolloids from Algal Sources

3.1.1. Agar

Agar is considered an excellent gel-forming polysaccharide and is mainly extracted from *Gracilariaceae* and *Gelidiaceae*, and particularly from species from *Gelidium*, *Gracilaria*, *Gracilariopsis*, *Pterocladia*, and *Gelidiella* genera [10].

Chemically, agar is composed of two main components: agarose, a linear polysaccharide built of repeating units of 1,3-linked β-D-galactopyranose (agarobiose units) and 1,4-linked 3,6-anhydro-α-L-galactopyranose (neoagarobiose units) with alternating sulfate groups; and agaropectin, a heterogeneous blend of smaller molecules characterized by a lower occurrence of alternating units of D-galactose and L-galactose, with acidic side chains, such as sulfate and pyruvate, or methoxyl groups [11].

Besides being extensively used as a solidifying and stabilizing component in the food industry [12], agar has also a unique potential in biomedical and pharmaceutical fields, such as described in Table 2.

3.1.2. Alginate

Alginate is a naturally occurring structural hydrocolloid widely distributed in the cell walls of brown seaweed (Phaeophyceae), such as *Laminaria hyperborea*, *Laminaria digitata*, *Laminaria japonica*, *Ascophyllum nodosum*, and *Macrocystis pyrifera*. As brown seaweed pigments are usually co-extracted, its color ranges from white to yellowish-brown [13].

Alginates are linear copolymers composed of block-like structures of β-D-mannuronic acid (M), and 1–4-linked α-L-guluronic (G) residues arranged in di-axial links, with a homogeneous (poly-G, poly-M) or heterogeneous (MG) block composition [14]. They are mainly used as slimming aids, thickening and/or gelling agents, as also in textile manufacturing, or waterproofing and fireproofing fabrics [15].

However, thanks to its good biocompatibility, non-toxic nature and lack of reactivity, alginates are also used in pharmaceutical and biomedical applications (Table 2).

3.1.3. Carrageenan

Carrageenan is a sulfated hydrocolloid with a structural role in red seaweeds, where it may correspond to as much as 30–70% of the seaweed's dry weight. Among red seaweed, carrageenan is mainly extracted from species such as *Chondrus crispus*, *Eucheuma cottonii*, and *Eucheuma spinosum* species [16].

The carrageenans with highest commercial relevance, are kappa (κ), iota (ι), and lambda (λ) forms. Kappa-carrageenan is particularly prevalent in species like *Kappaphycus alvarezii* and several *Eucheuma* species [17]; in turn, ι-carrageenan is primarily found in *Eucheuma denticulatum*, while λ-carrageenan is abundant in red seaweeds such as *Sarcopeltis skottsbergii* and *Chondrus crispus* [18,19].

Structurally, carrageenans are composed of linear chains made up of alternating 1,3-linked β-D-galactopyranose and 1,4-linked α-galactopyranose units. The κ, ι, and λ forms differ in the number and position of sulfate ester groups (one in κ-carrageenan, two in ι-carrageenan, and three in λ-carrageenan) and the presence of 3,6-anhydro-D-galactopyranose units [20,21].

Beyond food applications, carrageenan is widely utilized in pharmaceutical and cosmetic sectors, owing to its reported range of biological activities (Table 2).

Table 2. Main bioactivities and applications in pharmaceutical and biomedical fields of hydrocolloids from algae sources.

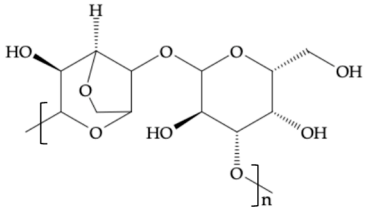
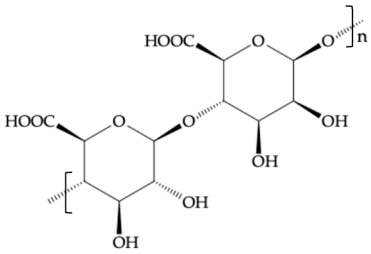
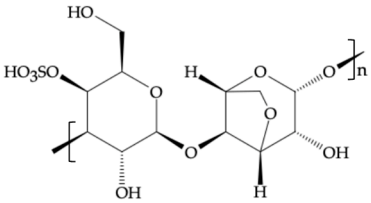
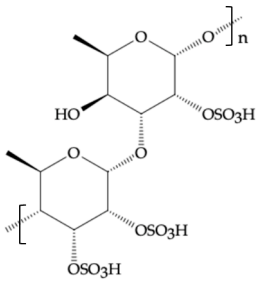
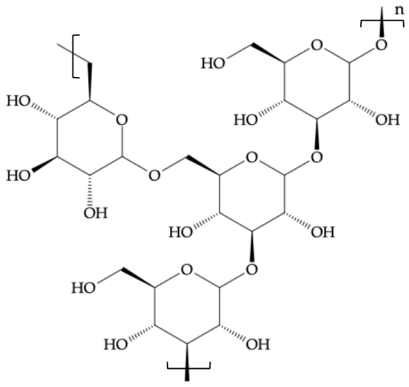
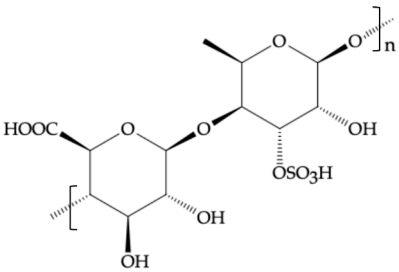
Component	Chemical Structure	Bioactivity/Biomedical Application
Agar		Hypoglycemic, anti-agglutinating activity, promotor of neurogenesis, angiogenesis, cartilage formation, and bone generation [22,23]. Agarose-based biomaterials are highly versatile in specific tissue development and engineering, drug delivery, wound repair, and fabrication of biological structure analogs [24,25].
Alginate		Application in tissue engineering, hydrogels, microspheres, nanoparticles, drug delivery matrices [26], prevention of acid reflux, and wound dressings [14,15,27].
κ-Carrageenan		Immunomodulatory, anticoagulant, antithrombotic, antiviral, and antitumor activity. Employed also in drug delivery systems and tissue regeneration through the targeted delivery of therapeutic biomacromolecules and cells [21,28,29].
Fucoidan		Anticoagulant, antithrombotic, antiviral, antitumor, immunomodulatory, antioxidant, and anti-inflammatory properties [30]. Previously used to treat infections, chronic inflammation, tissue injury, fibrosis, and neurological damage [31], and also in surgical and regenerative applications (due to its structural similarity to heparin) [32].

Table 2. *Cont.*

Component	Chemical Structure	Bioactivity/Biomedical Application
Laminarin		Strong anti-inflammatory and antioxidant activities, immunomodulatory, anti-angiogenic, and antitumoral (proapoptotic) effects [33]; combined with monoclonal antibodies and cytokines, it has also improved the results obtained in immunotherapy strategies [34]. Used also in tissue engineering, topical creams to treat wounds (accelerating collagen deposition and re-epithelization), protection against free radical-mediated oxidative stress and bacterial infections [35].
Ulvan		Ulvan demonstrated antioxidant, anti-inflammatory, immune-modulating, anticoagulant, antibacterial, antifungal, antiviral, cytotoxic, antitumor, anti-hyperlipidemic, and hepatoprotective effects [36]. It has also been incorporated as biomaterial for wound healing, tissue engineering, drug delivery system, and prebiotic formulations [37].

3.1.4. Fucoidan

Fucoidans are fucose-rich sulfated polysaccharides, primarily extracted from brown seaweed and marine invertebrates [38,39]. Within the cell walls of brown algae, fucoidans are the second most prevalent class of polysaccharides, potentially representing up to 20% of the organism's dry weight [40]. The most significant commercial sources include *Ascophyllum nodosum*, *Chorda filum*, *Cladosiphon okamuranas*, *Ecklonia kuroma*, *Fucus distichus*, *Fucus evanescens*, *Fucus vesiculosus*, *Laminaria japonica*, and *Undaria pinnatifida* [41].

Structurally, fucoidans comprise a broad class of fucans, predominantly composed of fucose (40–80%), sulfate groups, and uronic acids, with smaller percentages of other polysaccharides like algin and laminarin. Fucoidans are considered the most structurally heterogeneous polysaccharides (molecular weight varying among 100 to 5200 kDa) found in seaweed [42]. Its structure typically features a backbone of α-(1→3)-L-fucopyranose units or alternating α-(1→3) and α-(1→4)-linked L-fucopyranosyl residues. These residues may be further substituted with sulfate or acetate groups, or glycosyl units, such as glucuronic acid. It also contains small quantities of other monosaccharides like D-xylose, D-galactose, D-mannose, and uronic acids [43].

In recent years, fucoidan has gained considerable interest in biomedical and pharmaceutical research due to its diverse biological effects (Table 2).

3.1.5. Laminarin

Laminarin (or laminaran) is a biodegradable, non-toxic polysaccharide with significant potential across different applications [44]. It is one of the primary polymeric carbohydrates produced by brown algae and was initially isolated from their cell walls. Some of its most common sources include the species *Laminaria hyperborean*, *Laminaria digitata*, *Fucus vesiculosus*, and *Ascophyllum nodosum*, besides others belonging to the genera *Saccharina* and *Eisenia* [45].

Structurally, laminarin is a β-(1→3)-glucan with β-(1→6)-linked side chains, with the branching pattern varying depending on environmental conditions and harvesting season of the algae [33]. There are two types of polymeric chains within laminarin: M-chains, with mannitol at the reducing end, and G-chains, which terminate with a glucose unit [46].

Laminarin has attracted interest in the past years due to its promising results in diverse biomedical applications (Table 2).

3.1.6. Ulvan

Ulvan is found in green seaweeds (Chlorophyceae), particularly in temperate and tropical marine climates. Prominent sources of ulvan include *Ulva armoricana*, *Ulva clathrata*, *Ulva fasciata*, *Ulva ohnoi*, *Ulva intestinalis*, *Ulva lactuca*, *Ulva pertusa*, *Ulva rigida*, and *Ulva rotundata*. In addition to the *Ulva* genus, substantial concentrations of ulvan have also been identified in *Gayralia* and *Monostroma* species [37].

Ulvan is an anionic heteropolysaccharide [47] composed of varying sugar residues such as sulfated rhamnose, xylose, glucose, cellulose, xyloglucan, glucuronan, mannose, galactose, and arabinose, as well as uronic acids like glucuronic acid, or iduronic acid [48]. Its typical structural backbone consists of repeating disaccharide units of α-L-rhamnose-3-sulfate-1,4-β-D-glucuronic acid (ulvano-biuronic acid A) or α-L-rhamnose-3-sulfate-1,4-α-D-iduronic acid (ulvanobiuronic acid B); additionally, it also contains aldobiose sequences, referred to as ulvanobioses [49].

Ulvan has demonstrated diverse types of bioactivity, as detailed in Table 2.

3.2. Hydrocolloids from Animal Sources

3.2.1. Chitin

Chitin can be obtained from various natural sources, such as crustaceans, insects, cephalopod beaks, fish scales, and the skin of lissamphibians; it is also a primary structural component of fungal cell walls (e.g., *Aspergillus* and *Penicillium*) [50].

Chitin is made up of repeating units of N-acetylglucosamine (2-acetylamino-2-deoxy-D-glucose), being structurally similar to cellulose, as both consist of sugar units linked by β-(1→4) bonds; however, chitin presents an acetamide group in the C2 carbon instead of the hydroxyl group found in cellulose [51].

Chitin has been utilized, along with its byproducts, in different pharmaceutical applications (Table 3).

3.2.2. Chitosan

Chitosan is among the most abundant natural polymeric carbohydrates and is mainly obtained from the hard exoskeletons of arthropod shellfish, including crab, lobster, and shrimp. Commercially, it is typically produced through the deacetylation of chitin, which involves the alkaline hydrolysis of acetamide groups [52].

It consists of β-(1→4)-linked units of 2-acetylamino-2-deoxy-β-D-glucose and 2-amino-2-deoxy-β-D-glucose. The presence of both hydroxyl and amino groups allows chitosan to form covalent bonds through different chemical reactions, like esterification, reductive amination, and etherification reaction [53].

Besides its uses in cell encapsulation, food packaging, gelling and coating, developing active biopolymeric nanofilms [53], chitosan has demonstrated potential for a wide range of biomedical uses (Table 3).

3.2.3. Chondroitin Sulfate

This sulfated glycosaminoglycan is commonly found in the extracellular matrix of connective tissues, bone, cartilage, skin, ligaments, and tendons, in the form of proteoglycans [54]. Several marine organisms, such as sponges, sea cucumbers, squids, mollusks,

and cartilaginous fishes (e.g., shark, salmon, or ray) are considered valuable natural sources of chondroitin sulfate [55].

The basic unit of this complex glycosaminoglycan consists of a repeating disaccharide of glucuronic acid and N-acetyl-galactosamine linked by β-(1→3) bonds. These disaccharides are interconnected by β-(1→4) glycosidic bonds, forming a linear chain that reaches a molecular weight typically ranging between 5 and 50 kDa [54].

Chondroitin sulfate is recognized by the World Health Organization as an anti-inflammatory agent, being biomedically used for several purposes (Table 3).

3.2.4. Dermatan Sulfate

This galactosaminoglycan is typically present as a mucopolysaccharide in the skin, but it might also be found in other tissues, like lungs, heart, or blood vessels.

Similarly to chondroitin sulfate, it contains approximately equimolar proportions of hexosamine, acetyl groups, uronic acids, and sulfuric acids, besides sharing the same *N*-acetyl-D-galactosamine (GalNAc). However, they differ in the uronic acid component: dermatan sulfate contains L-iduronic acid (IdoA), while chondroitin sulfate contain D-glucuronic acid (GlcA). The repeating disaccharide units of dermatan sulfate are linked via alternating β-(1→4) glycosidic bonds, connecting GalNAc to IdoA, and α-(1→3) glycosidic bonds linking IdoA to GalNAc [56].

Dermatan sulfate had validated biomedical and pharmaceutical uses (Table 3).

Table 3. Main bioactivities and applications in pharmaceutical and biomedical fields of hydrocolloids from animal sources.

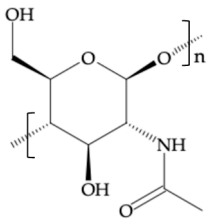
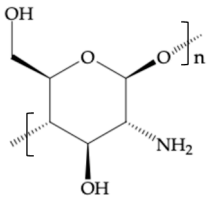
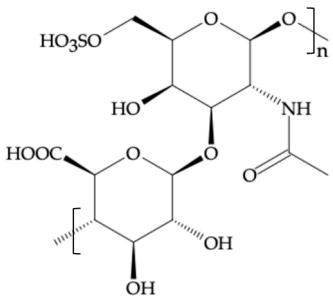
Component	Chemical Structure	Bioactivity/Biomedical Application
Chitin		As it is non-toxic, nonallergic, biodegradable, and insoluble in water, chitin has been used as a natural antimicrobial, promotor of non-specific immune responses, inhibitor of tumor cell proliferation, activator of peritoneal macrophages in vivo, and accelerator of wound healing process [57].
Chitosan		Capacity to enhance drug bioavailability, promoting gene therapy, wound dressing, and tissue engineering; also used in biosensors and bioimaging applications, cosmetic formulations for skin and hair care, renewable artificial skin, bone reconstruction, and different dentistry (e.g., dental implants) and ophthalmology (e.g., contact lenses, surgical sutures) treatments [52]. It has demonstrated therapeutic potential in areas such as cancer therapy, obesity management, and diabetes care [58].
Chondroitin sulfate		Important anti-inflammatory agent (listed as anti-inflammatory and non-steroidal anti-rheumatic by World Health Organization), particularly to treat osteoarthritis. It demonstrated capacity to modulate cell proliferation, differentiation, morphogenesis, and migration, prevent bacterial and viral infections. It has been incorporated into a diversity of pharmaceutical formulations, gene delivery systems, tissue engineering applications, and biomarkers for diagnosing different diseases [59].

Table 3. *Cont.*

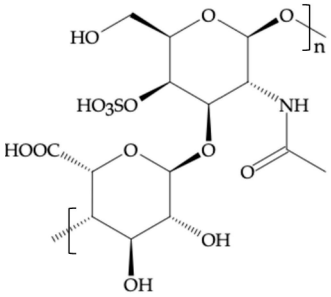
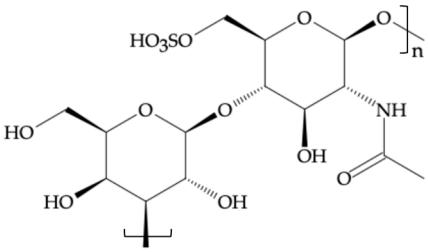
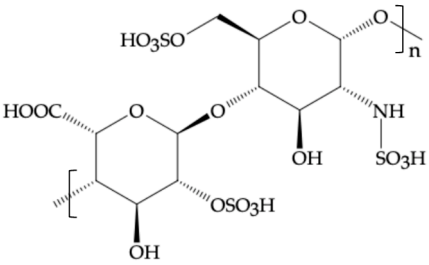
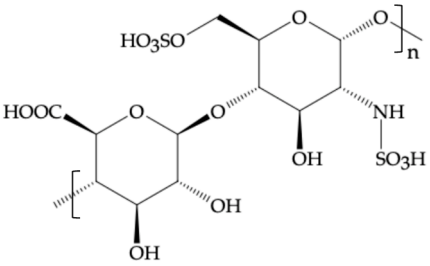
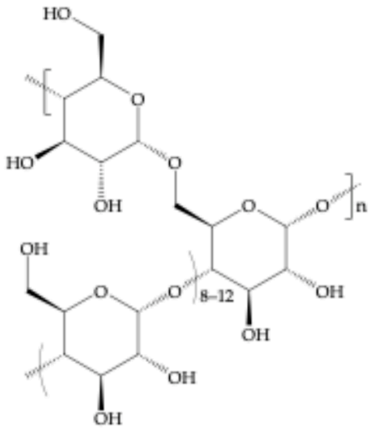
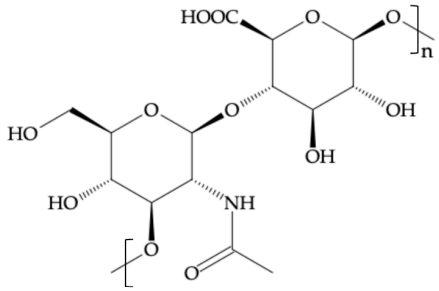
Component	Chemical Structure	Bioactivity/Biomedical Application
Dermatan sulfate		Antithrombotic agent that acts independently of antithrombin III, making it a promising alternative to heparin, and a possible adjuvant to treat certain vascular conditions. Dermatan sulfate might also be used in topical drug delivery systems and as a therapeutic agent in regenerative medicine and cancer treatment [60,61].
Keratan sulfate		Acknowledged as being useful to maintain corneal transparency, keratan sulfate also plays key roles in developmental processes, cell signaling, adhesion, and migration. Its pharmaceutical uses include the treatment of inflammatory conditions such as rheumatoid arthritis and chronic obstructive pulmonary disease [62,63].
Heparin		Heparin prevents the formation of new blood clots and limits the growth of existing ones [64]. Clinically, it is used to manage different conditions, including acute coronary syndrome, atrial fibrillation, deep-vein thrombolysis, pulmonary embolism, cardiopulmonary bypass during heart surgery, extracorporeal life support, and hemofiltration, although its use can trigger an immune-mediated side effect known as thrombocytopenia (leading to a reduction in platelet count) [65]. Besides its anticoagulation activity, heparin is used in other medical conditions such as cancer treatment, respiratory diseases, neurodegenerative disorders, pancreatitis, and rheumatoid arthritis [66].
Heparan sulfate		Acts as coreceptor on the cell surface, binding and modulating the bioactivity of specific ligands, and regulating processes such as cell proliferation, differentiation, migration, angiogenesis, and tissue morphogenesis and homeostasis [67,68]. It is an important therapeutic target and a functional biomaterial for tissue engineering, controlled drug delivery, or wound healing [69], besides participating in inflammatory responses [70].
Glycogen		Glycogen-based nanocomposites hold a significant interest in biomedical research, as these are involved in important physiologic processes, including calcium release from the sarcoplasmic reticulum, regulation of muscle excitability, potassium homeostasis, modulation of enzyme activities, involvement in gene expression, translational and post-translational processes, and inorganic phosphate sequestration [71].

Table 3. *Cont.*

Component	Chemical Structure	Bioactivity/Biomedical Application
Hyaluronan		Promotes cell viability and early tissue regeneration, being widely employed in conventional drug delivery systems (such as nanoparticles, liposomes, and hydrogels), mostly as a viscosity enhancer, stabilizer, and lubricant (particularly in eye drops, nasal sprays, and injectable solutions) [72,73]. Additionally, it is used in ophthalmic formulations for the sustained release of antibiotics and anti-inflammatory agents [74]. It is being used as a biomaterial in drug delivery vehicles such as nanoparticles, liposomes, and hydrogels [73], emerging as promising carriers for targeted cancer therapies [75].

3.2.5. Keratan Sulfate

Keratan sulfate is one of the most abundant glycosaminoglycans, accounting for nearly 65% of the total glycosaminoglycan content of the cornea, where it exists associated with proteins. It is also present in the central and peripheral nervous systems [76].

Structurally, keratan sulfate is composed of N-acetyl glucosamine, usually sulfated at the 6-*O* position, and linked via a β-(1→4) glycosidic bond to D-galactose, which can also be 6-*O*-sulfated, though this sulfation can differ depending on the tissue type, development stage, and species [77].

It has shown therapeutic potential to treat different conditions (Table 3).

3.2.6. Heparin

This glucosaminoglycan stands out for its high negative charge density, a characteristic that enhances its suitability for specific applications, such as cell encapsulation [78]. Heparin is primarily biosynthesized in the Golgi apparatus of mast cells, which are mainly located in the skin, liver, lungs, and intestines of higher animals [79].

It is a structurally complex polysaccharide, consisting of repeating sulfated disaccharide units that vary in their degree of sulfation, although presenting 2-*O*-sulfate-α-l-iduronic acid and 6-*O*-sulfate-*N*-sulfate-α-d-glucosamine linked by a 1,4-glycosidic bond as the most common unit [80].

Heparin is widely recognized for its several biological activities, having been used in different biomedical applications (Table 3).

3.2.7. Heparan Sulfate

This glycosaminoglycan is ubiquitously distributed, being more predominantly found on the surface of animal cells and within the extracellular matrix, where it is covalently attached to core proteins to form heparan sulfate proteoglycans [81].

Structurally, heparan sulfate consists of repeating disaccharide units of a uronic acid (alternating D-glucuronic acid and L-iduronic acid) linked β-(1→4) to a D-glucosamine residue. These disaccharides undergo variable N-acetylation or N-sulfation at the amino group of glucosamine, as well as *O*-sulfation (e.g., at the 2-*O*, 6-*O*, and sometimes 3-*O* positions), producing extensive heterogeneity in chain length and sulfation pattern [69], which enables diverse biological activities [82].

Heparan sulfate is a highly versatile and structurally complex glycosaminoglycan, whose biochemical heterogeneity underlies a broad spectrum of biological functions (Table 3).

3.2.8. Glycogen

This storage polysaccharide is found in mammalian and non-mammalian animal cells, mainly in the liver and the skeletal muscle, but also in other tissues such as brain, heart, kidney, adipose tissue, and erythrocytes, although its function in these tissues, where it is found in lower concentrations, remains largely unclear [83].

Chemically, its structure is related to that presented by starch, closely resembling amylopectin, as it consists of linear chains of D-glucopyranosyl units linked by α-(1→4) glycosidic bonds, with branching points formed through α-(1→6) bonds originating from 1,4,6-tri-*O*-substituted α-D-glucopyranosyl residues [84].

Glycogen has unique properties of biocompatibility and biodegradability, and specific glycogen-based nanocomposites are gaining interest in biomedical research (Table 3).

3.2.9. Hyaluronan

Hyaluronan, also known as hyaluronic acid, is a negatively charged polymeric carbohydrate extensively distributed throughout the extracellular matrix, as well as in epithelial, soft connective, and neuronal tissues [85].

This linear glycosaminoglycan is composed of repeating disaccharide units of β-D-N-acetylglucosamine and β-(1→4)-D-glucuronic acid [86].

In recent years, hyaluronan has drawn significant attention from the medical and pharmaceutical industries due to its versatile biological functions (Table 3).

3.3. Hydrocolloids from Plant Sources

3.3.1. Pectin

Pectin, together with cellulose and hemicellulose, is a key structural polysaccharide found in fruit skins. In commercial applications, it is mainly extracted from citrus fruits' peels, being considered as a highly valuable polymeric carbohydrate due to its significant viscosity and its relevance as a fundamental component of dietary fiber [87].

Structurally, pectin consists of a complex group of polysaccharides predominantly located in the primary cell walls of plants, especially in soft, non-woody tissues [88]. It is primarily composed of a backbone of α-(1→4)-linked polygalacturonic acid units, intercalated with rhamnose residues, with side chains containing neutral sugars. The carboxyl groups on the galacturonic acid units can be esterified with non-carbohydrate groups such as methyl or acetyl groups [89].

Beyond its applications in food, pectin is also incorporated in pharmaceutical formulations due to its biological activity (Table 4).

3.3.2. Starch

Starch is the most prevalent storage polysaccharide found in nature and is abundant in legumes, grains, and tuberous plants. Some of its main sources include rice, wheat, corn, and potatoes [90].

Structurally, starch consists of D-glucose units linked through α-(1→4) glycosidic bonds in linear chains and α-(1→6) at branched points. It is generally divided into amylose, which is mostly linear, and amylopectin, which is highly ramified; the pattern of the branching structure affects significantly the way pectin is digested and metabolized [91].

Although less extensively than other hydrocolloids, starch has also been used in biomedical applications (Table 4).

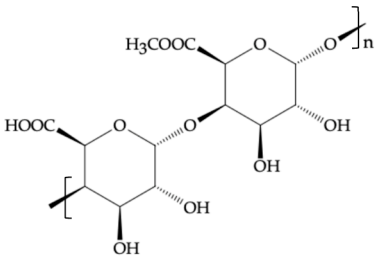
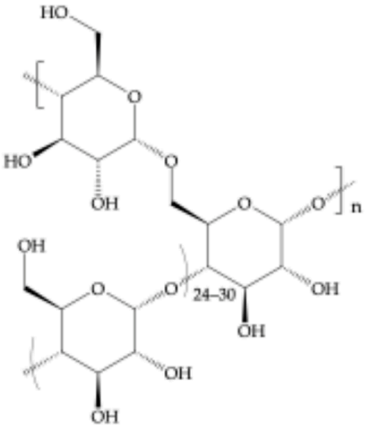
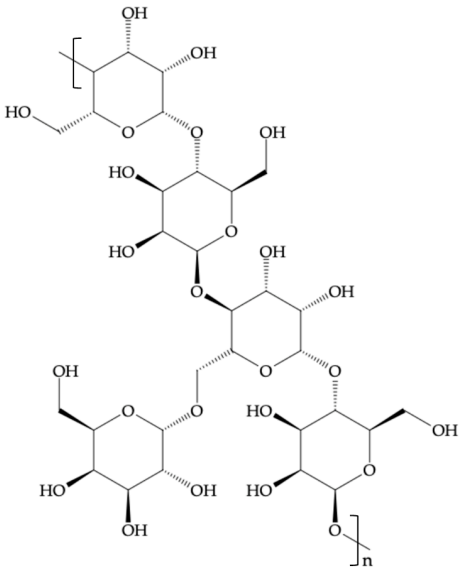
3.3.3. Locust Bean Gum

Locust bean gum is derived from the endosperm of locust (carob) seeds *Ceratonia siliqua* (L.) Taub (Leguminosae). Seed processing typically involves thermal, mechanical

or chemical treatment, germ elimination, milling and screening of the endosperm [92]. The resulting gum is a white to yellowish, nearly odorless, powder composed mainly of high-molecular-weight polysaccharides (50,000–3,000,000 Da), of which at least 75% are galactomannans. These galactomannans consist of a linear chain of (1→4)-linked β-D-mannopyranosyl units with (1→6)-linked α-D-galactopyranosyl side chains. Mannose and galactose contents range from 73 to 86% and 27 to 14%, respectively, resulting in a typical mannose:galactose ratio of about 4:1. Galactomannans are also found in other gums such as guar, tara or cassia, although with different mannose-to-galactose ratios [93].

Despite being more associated with food applications, there is some evidence of its use in pharmaceutical-related applications (Table 4).

Table 4. Main bioactivities and applications in pharmaceutical and biomedical fields of hydrocolloids from plant sources.

Component	Chemical Structure	Bioactivity/Biomedical Application
Pectin		Pectin has been combined with different polymers, enhancing its antioxidant capacity, suitability as bone substitute, tissue regeneration, anticancer potential, antimicrobial effects, and modulation of microbiota both in animals and humans [94]. These mixtures have been applied in drug delivery systems and surface modification of biomaterials [95].
Starch		Owing to its effective gelation and film-forming properties, the helical cavities within amylose and the height branch of amylopectin, starch is adequate to encapsulate core materials [96], having been used for drug delivery, micro- and nano-preparations of therapeutic drugs, and tablet production [97].
Locust bean gum		Remarkable antibacterial and antioxidant properties [98]. Validated application in controlled drug release systems, cellular hyperproliferation and antimicrobial wound therapies, as well as in the enhancement of the bioadhesivity and skin permeation of different active compounds [99,100].

4. Cellulose-Based Semisynthetic Hydrocolloids

Cellulose is an essential fibrous material found primarily in plants such as cotton, flax, jute, or hemp. It is especially abundant in the ligneous parts of plant, particularly in tree trunks [101]. In its pure form, cellulose is insoluble in both hot and cold water, due to strong intramolecular hydrogen bonding, and is indigestible to the human gastrointestinal tract [45]. However, cellulose can be chemically modified into esters or ether derivatives to make it hydrosoluble [102]. Examples of such water-soluble cellulose derivatives include hydroxypropylcellulose, hydroxypropylmethylcellulose, hydroxyethylcellulose, and sodium carboxymethylcellulose [103].

4.1. Carboxymethylcellulose (E466)

Carboxymethylcellulose is a linear, long-chain, water-soluble, anionic carbohydrate polymer that may be prepared by the reaction of monochloroacetic acid with alkali cellulose [104]. It is one of the most important and widely used hydrosoluble gums at the industrial level [105].

Its compatibility with other ingredients, safety profile, color retention capacity, stain-removal properties stains, stability and lubrication ability, make carboxymethylcellulose a prime candidate for chemical modification and incorporation in formulations as a moisturizer and emulsifier [106].

4.2. Microcrystalline Cellulose (E460)

Microcrystalline cellulose is a purified, partially depolymerized, form of cellulose obtained by treating cellulose with mineral acids, using hydrochloric acid to reduce its degree of polymerization. It can be produced by different methods, including reactive extrusion, enzyme-mediated processes, steam explosion, and acid hydrolysis [107].

This semisynthetic hydrocolloid is characterized by its high thermal stability, rigidity, crystallinity, and density, along with being odorless, renewable, mechanically strong, and relatively simple to produce. These features make it valuable across diverse sectors including food, cosmetics, packaging, and pharmaceuticals. In addition, the abundance of surface hydroxyl groups enables various chemical modifications, which can further tailor and improve its physical and chemical performance [108].

4.3. Hydroxypropylcellulose (E463)

Hydroxypropylcellulose is an ether derivative of cellulose in which some hydroxyl groups are hydroxypropylated through reaction with propylene oxide. This non-ionic semisynthetic hydrocolloid is known for its effective thickening, stabilizing, and film-forming abilities (its films remain flexible even without plasticizers and resist tackiness under humid conditions). It is soluble in water and organic solvents, offering versatile functional properties beyond its established role in oral drug formulations: it is widely used in the formulation of artificial tears, as a lubricant for artificial eyes, as a thickener, a low-level binder in tablets, as an emulsion stabilizer, or scaffold in tissue engineering [109,110].

Moreover, hydroxypropyl cellulose can be improved through different methods such as grafting, crosslinking, and other chemical modifications, which markedly improve its thermal stability, resistance to degradation, and overall biocompatibility [111].

4.4. Hydroxypropylmethylcellulose (E464)

Hydroxypropylmethylcellulose is another type of cellulose ether derivative, in which one or more hydroxyl groups in the cellulose ring are substituted with ether groups. It is a hydrophilic, biodegradable and biocompatible polymer, with a wide range of applications, such as a thickener, stabilizer and emulsifier in coatings and cosmetics. Compared to

other methyl cellulose derivatives, hydroxypropylmethylcellulose exhibits a higher organic solubility and greater thermal flexibility [112].

Its biocompatibility, biodegradability, and safety for oral use support applications also in mucoadhesive systems, ophthalmic formulations, and controlled-release dosage forms. It also shows high drug-loading capacity and compatibility with diverse active pharmaceutical ingredients, including bupropion hydrochloride, diclofenac sodium, and acetaminophen [113].

4.5. Ethylcellulose (E462)

Ethylcellulose is a partially *O*-ethylated cellulose ether derivative, prepared by the reaction of ethyl chloride with alkali cellulose. This semisynthetic hydrocolloid is insoluble in water, but soluble in a variety of solvents, and is widely used in food and pharmaceutical related applications [114]. Its ability to incorporate antimicrobial substances and reinforcing agents, together with its hydrophobicity, barrier-forming characteristics, and heat-resistance, makes EC films highly suitable in food packaging applications, besides being used as a taste masking agent [115].

Among ethylcellulose's applications, the biomedical field has received growing interest due to its biocompatibility and widespread availability, having been widely used in pharmaceutical formulations as a binder, coating agent of tablets and multiparticulates to achieve controlled drug release, or in hemocompatible nanocarriers to deliver different therapeutic agents [114,116].

5. Comparative Functional Properties and Structure–Property Relationship of Hydrocolloids

5.1. Comparative Bioactivity of Hydrocolloids

The physicochemical behavior and potential applications of hydrocolloids are strongly dependent on structural parameters such as monosaccharide composition, molecular weight, branching pattern, sulfation degree, charge density, and intermolecular interactions. These structural features influence the hydration capacity, viscosity, gelation behavior, mechanical stability, and bioactivity, ultimately determining the suitability of each hydrocolloid for food, pharmaceutical, biomedical, and material applications.

Table 5 comparatively summarizes the main bioactivities, their association with structural features, and major applications reported for the hydrocolloids discussed in this review. Overall, sulfated marine and animal-derived polysaccharides exhibit the most pronounced intrinsic bioactivities, particularly anticoagulant, antiviral, antioxidant, and immunomodulatory effects, whereas plant-derived and cellulose-based hydrocolloids are more frequently associated with physicochemical functionality, stabilization, and controlled-release applications.

Table 5. Comparative overview of the principal bioactivities and applications of the hydrocolloids discussed in this review.

Hydrocolloid	Principal Bioactivities Discussed in This Manuscript	Structural Features Associated with Bioactivity	Main Applications Discussed in Manuscript
Agar/Agarose	Biocompatibility and cell-supportive properties	Neutral galactan backbone enabling stable hydrogel formation	Microbiological media, scaffolds, hydrogel matrices
Alginate	Biocompatibility, mild antioxidant activity	Carboxyl-rich guluronic/mannuronic acid blocks enabling ionic gelation	Encapsulation, tissue engineering, food gels

Table 5. *Cont.*

Hydrocolloid	Principal Bioactivities Discussed in This Manuscript	Structural Features Associated with Bioactivity	Main Applications Discussed in Manuscript
κ-Carrageenan	Antiviral, antioxidant, immunomodulatory	Sulfated galactan structure and charge density	Food structuring, pharmaceutical gels, edible films
Fucoidan	Anticoagulant, antioxidant, anticancer, antiviral, immunomodulatory	High sulfate content and fucose-rich backbone	Drug delivery, anticancer systems, biomedical materials
Laminarin	Antioxidant, immunomodulatory, antimicrobial	β-(1→3)-glucan backbone with occasional β-(1→6) branching contributing to biological recognition and bioactivity	Functional foods, wound healing systems, biomedical and bioactive materials
Ulvan	Antioxidant, antimicrobial, immunomodulatory	Sulfated polysaccharide structure rich in rhamnose and uronic acids	Wound healing, tissue engineering, bioactive films
Chitin	Biocompatibility, antimicrobial and wound-healing potential	Acetamide-rich β-(1→4)-linked polysaccharide structure enabling hydrogen bonding and biointerface interactions	Biomedical materials, wound dressings, packaging and environmental application
Chitosan	Antimicrobial, antioxidant, mucoadhesive and wound-healing activity	Cationic amino groups and degree of deacetylation governing electrostatic interactions and bioactivity	Drug delivery, tissue engineering, edible coatings, antimicrobial films and biomedical systems
Chondroitin sulfate	Anti-inflammatory and chondroprotective activity	Sulfated glycosaminoglycan structure	Cartilage repair and regenerative medicine
Dermatan sulfate	Anticoagulant and regenerative properties	Sulfated disaccharide units interacting with growth factors	Regenerative biomaterials and therapeutic systems
Keratan sulfate	Cell signaling and tissue regenerative functions	Sulfated glycosaminoglycan structure	Ophthalmic and neural biomaterials
Heparin	Strong anticoagulant activity	Highly sulfated glycosaminoglycan structure	Antithrombotic formulations and drug delivery
Heparan sulfate	Cell signaling modulation and regenerative activity	Sulfated domains mediating protein interactions	Tissue engineering and biomimetic matrices
Glycogen	Biocompatibility and bioactive carrier potential	Highly branched glucose polymer structure	Nanocarriers and bioactive delivery systems
Hyaluronan	Anti-inflammatory activity, wound healing, tissue regeneration	High hydrophilicity and extracellular matrix-mimetic structure	Cosmetics, viscosupplementation, regenerative medicine
Pectin	Antioxidant and prebiotic potential	Degree of esterification and galacturonic acid content	Functional foods, delivery systems, food gels
Starch	Nutritional and encapsulation functionality	Amylose/amylopectin organization and retrogradation behavior	Food systems, biodegradable materials, encapsulation
Locust bean gum	Biocompatibility and stabilizing functionality	Galactomannan structure promoting synergistic interactions	Thickening systems and edible films
Carboxymethylcellulose	Biocompatibility and carrier functionality	Anionic cellulose ether structure	Drug delivery systems, coatings, stabilizers
Microcrystalline cellulose	Biocompatibility and reinforcement capability	Crystalline cellulose microdomains	Pharmaceutical excipients and composite materials

Table 5. *Cont.*

Hydrocolloid	Principal Bioactivities Discussed in This Manuscript	Structural Features Associated with Bioactivity	Main Applications Discussed in Manuscript
Hydroxypropylcellulose	Mucoadhesive and film-forming functionality	Hydroxypropyl substitution enabling thermoresponsive behavior	Ophthalmic formulations and delivery systems
Hydroxypropylmethylcellulose	Mucoadhesive and controlled-release functionality	Amphiphilic cellulose ether structure	Controlled drug release and food stabilization
Ethylcellulose	Barrier and controlled-release functionality	Hydrophobic ethyl substitution	Controlled-release coatings and packaging systems

5.2. Structure–Rheology–Gelation Relationship in Hydrocolloid Systems

As it can be inferred from Section 4, the functionality of hydrocolloids is strongly modulated by the relationship between their molecular structure and supramolecular network formation [7]. Besides those structural parameters such as molecular weight, branching degree, or sulfation pattern, charge density and monosaccharide composition influence can also influence intermolecular interactions (including hydrogen bonding, electrostatic attraction, ionic crosslinking, hydrophobic association, and chain entanglement) [117] that will further determine the viscoelastic response of hydrocolloid systems, which are characterized by rheological parameters such as storage modulus (G′), loss modulus (G″), apparent viscosity, and gel strength [118]. Alginate gels, for instance, are primarily formed through Ca^{2+}-mediated "egg-box" junctions between guluronic acid units, generating highly elastic networks with elevated G′ values [119]. In turn, carrageenan gelation depends on coil-to-helix transitions and cation-induced helix aggregation, while pectin gelation is strongly modulated by the degree of methyl esterification and calcium availability, whereas cellulose derivatives and chitosan mainly rely on hydrogen bonding and chain entanglement mechanisms that contribute to shear-thinning and viscoelastic behaviors [120].

In either case, establishing quantitative structure–rheology relationships is essential for the rational design of hydrocolloid systems with tailored textural, mechanical, and delivery properties.

Table 6 summarizes representative correlations between molecular structure, intermolecular interactions, gelation mechanisms, and rheological performance of major hydrocolloids, highlighting how structural parameters define viscoelastic behavior and gel strength.

Table 6. Correlation between molecular structure, intermolecular interactions, gelation mechanisms, and rheological properties of representative hydrocolloids.

Hydrocolloid	Structural Features	Intermolecular Interactions	Gelation Mechanisms	Rheological Behavior	Rheological/Gel Parameters	Reference
Agar/Agarose	Linear chains of β-D-galactose and 3,6-anhydro-L-galactose	Hydrogen bonding and helix aggregation	Coil-to-double-helix transition followed by aggregation of helices	Strong thermoreversible gels with dominant elastic behavior (G′ > G″)	High gel strength; low tan δ; gelation typically at 30–40 °C	[121]
Alginate	Linear copolymer of β-D-mannuronic acid (M) and α-L-guluronic acid (G) blocks	Ca^{2+}-mediated ionic crosslinking between G-blocks	"Egg-box" junction zone formation	Strong viscoelastic and shear-thinning behavior; highly elastic hydrogels	Gel stiffness and G′ increase with G-content and Ca^{2+} concentration	[122]
κ-Carrageenan	Sulfated galactan containing 3,6-anhydrogalactose	Hydrogen bonding, electrostatic interactions, and K^{+}-mediated helix aggregation	Coil-to-helix transition followed by aggregation of double helices	Strong thermoreversible elastic gels with dominant G′ and brittle texture	Gel strength strongly influenced by sulfate content and ionic environment	[123]

Table 6. *Cont.*

Hydrocolloid	Structural Features	Intermolecular Interactions	Gelation Mechanisms	Rheological Behavior	Rheological/Gel Parameters	Reference
Fucoidan	Sulfated fucose-rich polysaccharide with variable sulfation patterns and branching	Electrostatic interactions, hydrogen bonding, and polyelectrolyte associations	Ion-mediated formation of hydrated viscoelastic polymer networks	Pronounced shear-thinning and viscoelastic behavior in hydrogel systems	Rheological behavior strongly influenced by sulfation degree, molecular weight, and ionic environment	[124]
Laminarin	β-(1→3)-glucan backbone with occasional β-(1→6) branching	Hydrogen bonding and chain association	Limited intrinsic gelation ability; contributes to weak hydrated associative networks	Moderate viscosity enhancement and weak viscoelastic behavior in polysaccharide systems	Rheological response influenced by concentration and interactions with copolymers such as starch	[125]
Ulvan	Sulfated heteropolysaccharide rich in rhamnose, xylose, and uronic acids	Hydrogen bonding, electrostatic interactions, and polyelectrolyte associations	Formation of hydrated viscoelastic networks through ionic interactions and chain association	Shear-thinning behavior with weak-to-moderate viscoelasticity depending on extraction conditions and concentration	Viscosity, storage modulus (G′), and textural properties strongly influenced by molecular organization and extraction procedure	[126]
Pectin (LM/HM)	Linear α-(1→4)-D-galacturonic acid backbone with variable degree of methyl esterification and RG-I branching	Ca^{2+}-mediated ionic crosslinking (LM), hydrogen bonding and hydrophobic interactions (HM)	Ionotropic "egg-box" gelation for LM pectins; acid/sugar-induced junction zone formation for HM pectins	Viscoelastic gels exhibiting shear-thinning behavior and dominant elastic response (G′ > G″) after network formation	Gel stiffness and G′ increase with Ca^{2+} concentration, molecular weight, and lower degree of methoxylation; HM gelation strongly depends on pH and soluble solids	[127]
Chitin	Highly crystalline β-(1→4)-linked N-acetylglucosamine polymer	Extensive hydrogen bonding, fibrillar association, and crystalline packing	Formation of rigid fibrillar and nanofiber-based hydrated networks	Weak-to-moderate viscoelastic behavior with shear-dependent structural organization in nanofiber/hydrogel systems	Storage modulus (G′), viscosity, and mechanical rigidity strongly influenced by nanofiber organization, crystallinity, and concentration	[128]
Chitosan	Partially deacetylated β-(1→4)-linked glucosamine polymer with tunable charge density	Hydrogen bonding, electrostatic interactions, and chain entanglement	pH- and ion-induced physical gelation; formation of transient polymeric networks	Pronounced shear-thinning and viscoelastic behavior; transition from viscous to elastic response during gelation	G′ progressively exceeds G″ during gel formation; viscosity and gel strength increase with molecular weight and degree of deacetylation	[129]
Chondroitin sulfate	Sulfated glycosaminoglycan composed of repeating glucuronic acid and N-acetylgalactosamine disaccharides	Electrostatic interactions, hydrogen bonding, and hydration forces	Formation of hydrated viscoelastic hydrogel matrices through covalent and supramolecular crosslinking	Viscoelastic and shear-dependent hydrogel behavior with enhanced mechanical stability in composite systems	Storage modulus (G′), loss modulus (G″), gelation kinetics, and stress relaxation behavior strongly influenced by crosslink density and polymer composition	[130]
Dermatan sulfate	Sulfated glycosaminoglycan containing iduronic acid-rich repeating disaccharides	Electrostatic interactions, hydrogen bonding, and hydration forces	Formation of hydrated viscoelastic matrices, especially in collagen-based composite hydrogels	Moderate viscoelastic behavior contributing to matrix stiffness modulation and transport properties	Mechanical stiffness, transport behavior, and matrix microarchitecture influenced by dermatan sulfate incorporation	[131]

Table 6. *Cont.*

Hydrocolloid	Structural Features	Intermolecular Interactions	Gelation Mechanisms	Rheological Behavior	Rheological/Gel Parameters	Reference
Keratan sulfate	Sulfated glycosaminoglycan composed of galactose and N-acetylglucosamine units	Electrostatic interactions, hydration, and proteoglycan-mediated associations	Formation of highly hydrated viscoelastic extracellular matrices	Weak-to-moderate viscoelastic behavior associated with tissue mechanical reinforcement	Mechanical stiffening behavior influenced by keratan sulfate density and matrix organization	[132]
Heparin	Highly sulfated glycosaminoglycan with strong negative charge density	Electrostatic interactions, hydrogen bonding, and protein-mediated associations	Formation of hydrated viscoelastic networks through supramolecular and covalent crosslinking in hydrogel systems	Pronounced viscoelastic and shear-thinning behavior in biofunctional hydrogels	Storage modulus (G′), gelation kinetics, and stress relaxation strongly influenced by crosslinking density and heparin content	[133]
Heparan sulfate	Heterogeneously sulfated glycosaminoglycan with highly hydrated domains	Electrostatic interactions, hydrogen bonding, and growth-factor-mediated associations	Formation of hydrated bioactive matrices through polysaccharide-protein interactions	Weak-to-moderate viscoelastic hydrogel behavior with strong hydration characteristics	Dynamic moduli and matrix organization influenced by sulfation pattern and biomolecular interactions	[133]
Glycogen	Highly branched α-(1→4)/α-(1→6)-glucan polymer forming compact particulate structures	Hydrogen bonding, molecular crowding, and particle association	Formation of hydrated particulate dispersions rather than true continuous gels	Low-viscosity viscoelastic dispersions with concentration-dependent rheological behavior	Viscosity and viscoelasticity influenced by branching density, particle size, and concentration	[134]
Hyaluronan	Linear polysaccharide composed of glucuronic acid and N-acetylglucosamine repeating units	Hydrogen bonding, chain entanglement, and hydration interactions	Formation of highly hydrated viscoelastic network through physical entanglement and chemical crosslinking	Strong shear-thinning and viscoelastic behavior characteristic of hydrogel systems	Storage modulus (G′), viscosity, injectability, and stress relaxation behavior strongly dependent on molecular weight and crosslinking density	[135]
Starch	Mixture of linear amylose and highly branched amylopectin chains linked by α-(1→4) and α-(1→6) glycosidic bonds	Hydrogen bonding, chain entanglement, and recrystallization interactions	Gelatinization upon heating followed by retrogradation during cooling	Pronounced shear-thinning and viscoelastic behavior; elastic gel formation after cooling	Higher amylose content generally increases G′, gel firmness, and retrogradation tendency	[136,137]
Locust bean gum	Galactomannan composed of a β-(1→4)-mannan backbone with α-(1→6)-galactose side chains	Hydrogen bonding and intermolecular chain entanglement; synergistic interactions with other hydrocolloids	Entanglement-driven thickening and synergistic gelation with carrageenan or xanthan gum	High viscosity and pronounced shear-thinning behavior	Synergistic blends exhibit significant increases in G′ and network strength	[138]
Cellulose derivatives	Chemically substituted cellulose ethers with tunable charge density and hydrophilic/hydrophobic balance	Hydrogen bonding, hydrophobic association, electrostatic interactions, and chain entanglement	Thermal or pH-induced formation of transient viscoelastic polymer networks	Pronounced non-Newtonian shear-thinning behavior with viscoelastic liquid-to-gel transition (G′ exceeding G″ during gelation)	Rheological properties strongly depend on substitution degree, temperature, pH, ionic strength, and polymer concentration	[139]

6. Conclusions

Hydrophilic colloids comprise a highly diverse group of natural and semisynthetic polysaccharides whose structural variability determines their physicochemical behavior and multipurpose bioactivity. This review highlights their performance in different sectors, but with a clear emphasis in pharmaceutical and biomedical applications, and how these are defined by a structure–property relationship involving molecular weight, monosaccharide composition, glycosidic linkages, sulfation pattern, and charge density.

Marine hydrocolloids such as alginate, carrageenan, fucoidan, ulvan, agar, and laminarin exhibit particularly versatile rheological and bioactive properties due to the presence of charged functional groups and ion-mediated interactions. Animal-derived polysaccharides, including chitosan, hyaluronan, chondroitin sulfate, heparin, and related glycosaminoglycans, demonstrate remarkable biological functionality and biocompatibility, supporting their growing use in drug delivery, regenerative medicine, tissue engineering, and bioactive hydrogel systems. Plant-derived hydrocolloids such as pectin, starch, and locust bean gum remain especially important due to their abundance, biodegradability, thickening ability, and synergistic interactions with other polymers, whereas cellulose-derived semisynthetic hydrocolloids provide tunable physicochemical performance and formulation versatility for industrial and pharmaceutical applications.

Importantly, the comparative analysis developed in this review demonstrates that the rheological behavior and gelation performance of hydrocolloids are direct consequences of their supramolecular organization and intermolecular interactions. Parameters such as hydrogen bonding, ionic crosslinking, electrostatic attraction, hydrophobic association, and chain entanglement critically influence viscosity, viscoelasticity, gel strength, stability, and controlled-release behavior. Consequently, understanding the relationship between molecular architecture and macroscopic functionality is essential for the rational design of advanced hydrocolloid-based systems.

Overall, the present review moves beyond a purely descriptive overview by integrating structural, rheological, and functional perspectives across different hydrocolloid families. This comparative approach provides a broader understanding of the factors governing hydrocolloid performance and may support the future development of sustainable, multifunctional, and application-oriented materials for food, pharmaceutical, biomedical, and technological fields.

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